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United States Patent Application Publication

20250271073

Kind Code

A1

Publication Date

August 28, 2025

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ASSEMBLY STRUCTURE OF VALVE SEAT

Abstract

Disclosed is a valve seat assembly structure capable of simply assembling a valve seat to a valve housing, including an inner wall and a single pair of first fastening portions having a shape that protrudes from the inner wall and provided to face each other; and a valve seat provided between the single pair of first fastening portions and including a second fastening portion formed on the side to couple with the first fastening portion, and each of the first fastening portion and the second fastening portion include an insertion shaped portion having a curved shape and a stopper shaped portion having a planar shape.

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Appl. No.: 18/983662

Filed: December 17, 2024

Foreign Application Priority Data

KR 10-2024-0028959

Feb. 28, 2024

KR 10-2024-0156423

Nov. 06, 2024

Publication Classification

Int. Cl.: F16K25/00 (20060101); F16K11/085 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority from and the benefit of Korean Patent Application No. 10-2024-0028959 filed on Feb. 28, 2024, and Korean Patent Application No. 10-2024-0156423 filed on Nov. 6, 2024, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Field

[0002] Example embodiments relate to an assembly structure of a valve seat, and more particularly, to a valve seat assembly structure that allows a valve seat to be easily assembled to a valve housing.

2. Related Art

[0003] In general, a valve is formed in a system that guides a flow direction of fluid. The valve is rotatably accommodated in an inner space of a valve housing. Here, a valve seat is provided between the valve housing and the valve.

SUMMARY

[0004] Example embodiments propose a structure that may stably assembly a valve seat to a valve housing.

[0005] According to an example embodiment, there is provided a valve seat assembly structure including a valve housing including an inner wall and a single pair of first fastening portions having a shape that protrudes from the inner wall and provided to face each other; and a valve seat provided between the single pair of first fastening portions and including a second fastening portion formed on the side to couple with the first fastening portion, wherein each of the first fastening portion and the second fastening portion includes an insertion shaped portion having a curved shape and a stopper shaped portion having a planar shape.

[0006] According to an aspect, the first fastening portion may include a (1-1)-th insertion shaped portion that vertically extends in the vertical direction of the valve housing; and a (1-1)-th stopper shaped portion that is formed in the inner direction of the first fastening portion at the end of the largest horizontal cross-section of the (1-1)-th insertion shaped portion.

[0007] According to an aspect, the first fastening portion may include a (1-2)-th insertion shaped portion that extends in a curved shape from the lower side of the (1-1)-th insertion shaped portion; and a (1-2)-th stopper shaped portion that extends in the inner direction of the first fastening portion at the end of the largest horizontal cross-section of the (1-2)-th insertion shaped portion.

[0008] According to an aspect, the second fastening portion may include a (2-1)-th insertion shaped portion that is formed in a straight area on the side of the valve seat, and vertically extends in the vertical direction of the valve seat; and a (2-1)-th stopper shaped portion that is formed in the inner direction of the second fastening portion at the end of the largest horizontal cross-section of the (2-1)-th insertion shaped portion.

[0009] According to an aspect, the second fastening portion may include a (2-2)-th insertion shaped portion that is formed in a curved area on the side of the valve seat and extends in a curved shape from the lower side of the (2-1)-th insertion shaped portion; and a (2-2)-th stopper shaped portion that extends in the inner direction of the second fastening portion at the end of the largest horizontal cross-section of the (2-1)-th insertion shaped portion.

[0010] According to an aspect, the valve seat may be curved on both left and right sides to have a constant curvature based on a vertical centerline.

[0011] According to an aspect, the valve seat may include a first wall portion formed on its outside;

and a second wall portion formed to surround a center hole of the valve seat, and an insertion groove may be formed between the first wall portion and the second wall portion.

[0012] According to an aspect, a sealing portion may insert into the insertion groove.

[0013] According to an aspect, the sealing portion may be an X-ring or an O-ring.

[0014] According to an example embodiment, there is provided a valve seat assembly structure including a valve housing including an inner wall and a single pair of first fastening portions having a shape that protrudes from the inner wall and provided to face each other; a valve seat formed with an insertion groove, provided between the single pair of first fastening portions, and having a second fastening portion to couple with the first fastening portion; and a sealing portion configured to insert into the insertion groove, wherein the valve seat includes a closed portion, a first wall portion formed on the outside based on the insertion groove, and a second wall portion formed on the inner side than the first wall portion, a chamfer portion obliquely provided to face the first fastening portion is formed in a portion in which an outer portion of the closed portion and an outer portion of the first wall portion are connected to each other, and a round portion having a first curvature is formed in a portion in which the outer portion of the closed portion and an outer portion of the second wall portion are connected to each other.

[0015] According to example embodiments, a valve seat may be easily assembled to a valve housing using a snap-fit method.

[0016] Also, according to example embodiments, when assembling a valve seat, a sealing portion accommodated in the valve seat may be stably and closely adhered to a valve housing.

[0017] Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Embodiments will be described in more detail with regard to the figures, wherein like reference numerals refer to like parts throughout the various figures unless otherwise specified, and wherein:

[0019] FIG. 1 illustrates a state in which a valve seat is mounted to a valve housing according to an example embodiment;

[0020] FIG. 2 illustrates a state in which a valve seat is partially separated from a valve housing according to an example embodiment;

[0021] FIG. 3 is a view cut along line A-A' of FIG. 2;

[0022] FIG. 4 is an enlarged plan view of a portion X of FIG. 2;

[0023] FIG. 5 is a front view of a valve seat according to an example embodiment;

[0024] FIG. 6 is a rear view of a valve seat according to an example embodiment;

[0025] FIG. 7 is a top view of a valve seat according to an example embodiment;

[0026] FIG. 8 illustrates a state in which a sealing portion is separated from a valve seat according to an example embodiment;

[0027] FIG. 9 illustrates a state in which a valve seat is mounted to a valve housing according to another example embodiment;

[0028] FIG. 10 illustrates a state in which a valve seat is partially separated from a valve housing according to another example embodiment;

[0029] FIG. 11 is an enlarged view of a portion Y of FIG. 10;

[0030] FIG. 12 is an enlarged view of a portion Z of FIG. 10;

[0031] FIG. 13 is a front view of a valve seat according to another example embodiment;

[0032] FIG. 14 is a rear view of a valve seat according to another example embodiment;

[0033] FIG. **15** is a view cut along line C-C' of FIG. **13**;

[0034] FIG. **16** illustrates a state in which a sealing portion is separated from a valve seat according to another example embodiment; and

[0035] FIG. **17** is a view cut along line B-B' of FIG. **10**.

DETAILED DESCRIPTION

[0036] Hereinafter, an assembly structure of a valve seat according to an example embodiment is described with reference to the accompanying drawings. Here, like reference numerals refer to like elements throughout, and repetitive description and detailed description related to known functions and configurations that may unnecessarily obscure the gist of the invention are omitted.

Implementation of the invention is provided to more fully explain the present invention for one of ordinary skill in the art. Therefore, shapes and sizes of elements in the drawings may be exaggerated for more clear explanation.

[0037] FIG. **1** illustrates a state in which a valve seat is mounted to a valve housing according to an example embodiment, FIG. **2** illustrates a state in which a valve seat is partially separated from a valve housing according to an example embodiment, FIG. **3** is a view cut along line A-A' of FIG. **2**, and FIG. **4** is an enlarged plan view of a portion X of FIG. **2**.

[0038] Referring to FIG. **1**, a valve seat assembly structure according to an example embodiment relates to a structure in which a valve seat **30** is assembled to a valve housing **10**.

[0039] The valve housing **10** may have an internal space such that the valve **20** may be rotatably accommodated. Here, the valve **20** may have a substantially cylindrical shape and may include a plurality of flow paths (T). A shaft **22** may be formed in the valve **20**. The shaft **22** may be formed in the vertical direction at the center of the valve **20**, and may have both ends that protrude outward along the vertical direction of the valve **20**. One end of the shaft **22** may be connected to an actuator (not shown) such that the valve **20** may rotate and another end of the shaft **22** may insert into an inserting portion **12** formed in a bottom **18** of the valve housing **10**.

[0040] A connecting hole **13** connectable to a port may be formed in an inner wall **19** of the valve housing **10**. A plurality of connecting holes **13** may be formed in the inner wall **19** of the valve housing **10** along the circumferential direction thereof. The flow path (T) formed in the valve **20** may be in communication with the connecting hole **13**. The valve seat **30** may be formed around the connecting hole **13** such that the valve **20** may be safely received. A detailed shape of the valve seat **30** is described below.

[0041] Referring to FIGS. **2** to **4**, the valve housing **10** may include a first fastening portion **15** such that the valve seat **30** may be fastened. The first fastening portion **15** may have a shape that protrudes from the inner wall **19** of the valve housing **10** toward the center of the valve housing **10**, and may be formed to extend along the vertical direction of the valve housing **10**. The first fastening portions **15** may be formed at a certain distance on both sides of the valve seat **30** to face each other. Here, two first fastening portions **15** formed to face each other may constitute a single pair. A plurality of pairs of such first fastening portions **15** may be formed on the inner wall **19** of the valve housing **10** along the circumferential direction thereof.

[0042] The first fastening portion **15** may include a first insertion shaped portion **15-11**, **15-12** and a first stopper shaped portion **15-21**, **15-22**. The first insertion shaped portion **15-11**, **15-12** may have a shape that helps the valve seat **30** insert into the first fastening portion **15**, and the first stopper shaped portion **15-21**, **15-22** may have a shape that helps the valve seat **30** insert into first fastening portion **15** and be stopped.

[0043] The first insertion shaped portion **15-11**, **15-12** may have an outer surface formed in the direction toward the inside of the valve housing **10**, and the first stopper shaped portion **15-21**, **15-22** may have an outer surface formed in a direction toward the inner wall **19** of the valve housing **10**.

[0044] In detail, referring to FIG. **2**, the first insertion shaped portion **15-11**, **15-12** may include a curved shape based on a horizontal cross-section of the first fastening portion **15**, and may have a

shape of which horizontal cross-sectional area gradually increases with getting closer to the inner wall **19** of the valve housing **10**. The curved shapes of the first insertion shaped portions **15-11**, **15-12** constituting a single pair may be formed to face each other. The first stopper shaped portion **15-21**, **15-22** may include a planar shape that extends from one end of the first insertion shaped portion **15-11**, **15-12** in the inner direction of the first fastening portion **15**.

[0045] The first insertion shaped portion **15-11**, **15-12** may include the (1-1)-th insertion shaped portion **15-11** that vertically extends in the vertical direction and the (1-2)-th insertion shaped portion **15-12** that extends in a curved shape from the lower end of the (1-1)-th insertion shaped portion **15-11** when the first fastening portion **15** is viewed from the front. The first stopper shaped portion **15-21**, **15-22** may include the (1-1)-th stopper shaped portion **15-21** that vertically extends in the vertical direction and the (1-2)-th stopper shaped portion **15-22** that extends in a curved shape from the lower end of the (1-1)-th stopper shaped portion **15-21** when the first fastening portion **15** is viewed from the front. Here, the (1-2)-th insertion shaped portion **15-12** and the (1-2)-th stopper shaped portion **15-22** present on both sides based on the valve seat **30** may have the curved shapes that approach each other.

[0046] Based on a direction from the center of the valve housing **10** toward the inner wall **19**, a distance between a single pair of first insertion shaped portions **15-11**, **15-12** that face each other may gradually decrease and a distance between the first fastening portions **15** may increase again from the first stopper shaped portion **15-21**, **15-22**.

[0047] FIG. **5** is a front view of a valve seat according to an example embodiment, FIG. **6** is a rear view of a valve seat according to an example embodiment, and FIG. **7** is a top view of a valve seat according to an example embodiment.

[0048] Referring to FIGS. **5** to **7**, the valve seat **30** may have a ring shape that surrounds a center hole **32**. As shown in FIG. **5**, in an example embodiment, the valve seat **30** may include a first area (**A1**) and a second area (**A2**). The first area (**A1**) may be a straight area formed on each of both left and right sides of the valve seat **30**, and the second area (**A2**) may be a curved area that extends upward or downward from each of both left and right sides of the valve seat **30**.

[0049] As shown in FIG. **5**, a contact portion **33** that may contact the valve **20** may be formed in the valve seat **30**. Here, the valve seat **30** may be formed to have a certain curvature to contact the outer peripheral surface of the valve **20**. Referring to FIGS. **5** and **7**, the valve seat **30** may be formed to be curved on both left and right sides with respect to a centerline (**C**). Here, the curved curvature of the valve seat **30** and the curvature of the outer peripheral surface of the valve **20** may be formed to be identical or similar to each other.

[0050] As shown in FIG. **6**, the valve seat **30** may include a first wall portion **36** and a second wall portion **37**. The first wall portion **36** may be formed on the outer end of the valve seat **30**, and the second wall portion **37** may be formed inside the valve seat **30** at a certain distance from the first wall portion **36**. The first wall portion **36** may form the outer surface of the valve seat **30**, and the second wall portion **36** may be formed to surround the center hole **32**.

[0051] An insertion groove **38** may be formed between the first wall portion **36** and the second wall portion **37**. The insertion groove **38** may have a ring shape along the first wall portion **36** and the second wall portion **37**. The upper side of the insertion groove **38** may be open and the opposite side of its bottom may be the contact portion **33** described above.

[0052] A second fastening portion **35** may be formed on each of both left and right sides of the valve seat **30**. The second fastening portion **35** may have a shape that protrudes outward from each of both left and right sides of the valve seat **30**, and may be formed to extend along the outer surface of the valve seat **30** in the vertical direction of the valve seat **30**.

[0053] The second fastening portion **35** may include a second insertion shaped portion **35-11**, **35-12** and a second stopper shaped portion **35-21**, **35-22**.

[0054] The second insertion shaped portion **35-11**, **35-12** may have the outer surface formed toward the rear of the valve seat **30**, and the second stopper shaped portion **35-21**, **35-22** may have

the outer surface formed toward the front of the valve seat **30**.

[0055] In detail, referring to FIGS. **5** to **7**, the second insertion shaped portion **35-11**, **35-12** may include a curved shape based on a horizontal cross-section of the second fastening portion **35**, and may have a shape of which the horizontal cross-section gradually increases with getting closer from the rear to the front of the valve seat **30**. The second stopper shaped portion **35-21**, **35-22** may include a planar shape that extends from one end of the second insertion shaped portion **35-11**, **35-12** in the inner direction of the second fastening portion **35**.

[0056] The second insertion shaped portion **35-11**, **35-12** may include the (2-1)-th insertion shaped portion **35-11** that vertically extends in the vertical direction and the (2-2)-th insertion shaped portion **35-12** that extends in a curved shape from the lower end of the (2-1)-th insertion shaped portion **35-11** when the second fastening portion **35** is viewed from the front. The second stopper shaped portion **35-21**, **35-22** may include the (2-1)-th stopper shaped portion **35-21** that vertically extends in the vertical direction and the (2-2)-th stopper shaped portion **35-22** that extends in a curved shape from the lower end of the (2-1)-th stopper shaped portion **35-21** when the second fastening portion **35** is viewed from the front.

[0057] The (2-1)-th insertion shaped portion **35-11** and the (2-1)-th stopper shaped portion **35-21** may be formed in the first area (A1) of the valve seat **30**, and the (2-2)-th insertion shaped portion **35-12** and the (2-2)-th stopper shaped portion **35-22** may be formed in the second area (A2) that is formed on the lower side of the valve seat **30**.

[0058] The (2-1)-th insertion shaped portion **35-11** and the (2-1)-th stopper shaped portion **35-21** may correspond to the (1-1)-th insertion shaped portion **15-11** and the (1-1)-th stopper shaped portion **15-21** of the first fastening portion **15**, respectively, and the (2-2)-th insertion shaped portion **35-12** and the (2-2)-th stopper shaped portion **35-22** may correspond to the (1-2)-th insertion shaped portion **15-12** and the (1-2)-th stopper shaped portion **15-22** of the first fastening portion **15**, respectively. Here, when the first fastening portion **15** and the second fastening portion **35** are viewed from the front, the curvature of the curved shape of the (2-2)-th insertion shaped portion **35-12** and the (2-2)-th stopper shaped portion **35-22** may be formed to be identical or similar to the curvature of the curved shape of the (1-2)-th insertion shaped portion **15-12** and the (1-2)-th stopper shaped portion **15-22** of the first fastening portion **15**.

[0059] Meanwhile, a distance between portions that protrude most from both left and right sides of the valve seat **30** in the second insertion shaped portion **35-11**, **35-12** may be greater than a distance between portions that are formed to face each other and protrude most from both left and right sides in the first insertion shaped portion **15-11**, **15-12** among the pair of first fastening portions **15**.

[0060] FIG. **8** illustrates a state in which a sealing portion is separated from a valve seat according to an example embodiment.

[0061] Referring to FIG. **8**, in an example embodiment, a sealing portion **40** may insert into the insertion groove **38** of the valve seat **30**. The sealing portion **40** may be formed of a rubber material. The sealing portion **40** may have a ring shape. For example, the sealing portion **40** may have a shape of X-ring and O-ring.

[0062] According to an example embodiment, while the first fastening portion **15** of the valve seat **30** and the second fastening portion **35** of the valve seat **30** couple with each other, the valve seat **30** may be fastened to the valve housing **10** using a snap-fit method.

[0063] Referring to FIGS. **2** to **4**, when the valve seat **30** is provided to be assembled on the inner wall **19** of the valve housing **10**, the first insertion shaped portion **15-11**, **15-12** of the first fastening portion **15** and the second insertion shaped portion **35-11**, **35-12** of the second fastening portion **35** may be provided to have the curved shape in opposite directions. The second insertion shaped portion **35-11**, **35-12** may move along the outer surface of the first insertion shaped portion **15-11**, **15-12**. Here, the first fastening portion **15** may be elastically deformed to increase the distance between a single pair of first fastening portions **15** or the second fastening portion **35** may be elastically deformed to decrease the distance between the second fastening portions **35** formed on

left and right sides. When the end of the second insertion shaped portion **35-11**, **35-12** reaches the end of the first insertion shaped portion **15-11**, **15-12** (e.g., end with a largest area in the horizontal cross-section) and moves further, the second fastening portion **35** may be located on the inner side between the first fastening portions **15** and the first fastening portion **15** or the second fastening portion **35** may be elastically restored. Here, when the first stopper shaped portion **15-21**, **15-22** and the second stopper shaped portion **35-21**, **35-22** are provided to face each other and move in a direction opposite to an insertion direction of the valve seat **30**, the valve seat **30** may be fastened to a single pair of first fastening portions **15** as the first stopper shaped portion **15-21**, **15-22** and the second stopper shaped portion **35-21**, **35-22** come into contact with each other.

[0064] Meanwhile, only the (1-1)-th insertion shaped portion **15-11** and the (1-1)-th stopper shaped portion **15-21** may be formed in the first fastening portion **15**, and the (1-2)-th insertion shaped portion **15-12** and the (1-2)-th stopper shaped portion **15-22** may not be formed in the first fastening portion **15**.

[0065] FIG. **9** illustrates a state in which a valve seat is mounted to a valve housing according to another example embodiment, FIG. **10** illustrates a state in which a valve seat is partially separated from a valve housing according to another example embodiment, FIG. **11** is an enlarged view of a portion Y of FIG. **10**, and FIG. **12** is an enlarged view of a portion Z of FIG. **10**.

[0066] Referring to FIG. **9**, a valve seat assembly structure according to another example embodiment relates to a structure in which a valve seat **300** is assembled to a valve housing **100**.

[0067] Meanwhile, in the valve seat assembly structure according to another example embodiment, description related to the valve **20**, a connecting hole **130** formed in an inner wall **190** of the valve housing **100**, and an inserting portion **120** formed in a bottom **130** may refer to description made above in the valve seat assembly structure according to an example embodiment.

[0068] Referring to FIGS. **10** to **12**, the valve housing **100** may include a first fastening portion **150** such that the valve seat **300** may be fastened. The first fastening portion **150** may have a shape that protrudes from the inner wall **190** of the valve housing **100** toward the center of the valve housing **100**, and may be formed to extend along the vertical direction of the valve housing **100**. The first fastening portions **150** may be formed at a certain distance on both sides of the valve seat **300** to face each other. Here, two first fastening portions **150** formed to face each other may constitute a single pair. A plurality of pairs of such first fastening portions **150** may be formed on the inner wall **190** of the valve housing **100** along the circumferential direction thereof.

[0069] The first fastening portion **150** may include a first insertion shaped portion **150-1** and a first stopper shaped portion **150-2**. The first insertion shaped portion **150-1** may have a shape that helps the valve seat **300** insert into the first fastening portion **150**, and the first stopper shaped portion **150-2** may have a shape that helps the valve seat **300** insert into the first fastening portion **150** and be stopped.

[0070] The first insertion shaped portion **150-1** may have an outer surface formed in a direction toward the inside of the valve housing **100**, and the first stopper shaped portion **150-2** may have a shape formed in a direction toward the inner wall **190** of the valve housing **100**.

[0071] In detail, the first insertion shaped portion **150-1** may have a curved shape based on a horizontal cross-section of the first fastening portion **150**, and may have a shape of which horizontal cross-sectional area gradually increases with getting closer to the inner wall **190** of the valve housing **100**. The curved shapes of the first insertion shaped portions **150-1** constituting a single pair may be formed to face each other. The first stopper shaped portion **150-2** may include a planar shape that extends from one end of the first insertion shaped portion **150-1** in the inner direction of the first fastening portion **150**.

[0072] The first insertion shaped portion **150-1** may have a shape that extends along the vertical direction. The first insertion shaped portion **150-1** may have a shape that extends such that its upper end and lower end have the identical or similar horizontal cross-section, or extends such that the horizontal cross-section gradually increases with getting closer from the upper side to the lower

side.

[0073] Based on a direction from the center of the valve housing **100** toward the inner wall **190**, a distance between a single pair of first insertion shaped portions **150-1** that face each other may gradually decrease and a distance between the first fastening portions **150** may increase again from the first stopper shaped portion **150-2**.

[0074] Referring to FIG. **11**, the first fastening portion **150** may include a first face **150-3** that extends from another end of the first stopper shaped portion **150-2** toward the inner wall **190** of the valve housing **100** and a second face **150-4** that extends from a protruding end of the first insertion shaped portion **150-1** toward the inner wall **190** of the valve housing **100**. A distance between the first face **150-3** and the second face **150-4** may be formed to be farther away from each other based on a direction toward the inner wall **190** of the valve housing **100**. When a single pair of first fastening portions **150** that face each other are viewed from above, the first faces **150-3** may be formed in a parallel posture with respect to each other, and the second faces **150-4** may be formed in a radial posture based on the center of valve housing **100**.

[0075] FIG. **13** is a front view of a valve seat according to another example embodiment, FIG. **14** is a rear view of a valve seat according to another example embodiment, and FIG. **15** is a view cut along line C-C' of FIG. **13**.

[0076] Referring to FIGS. **13** to **15**, the valve seat **300** may have a ring shape that surrounds the center hole **320**. The valve seat **300** according to another example embodiment may include a first area (A1') and a second area (A2'). The first area (A1') may be a straight area formed on each of both left and right sides of the valve seat **300**, and the second area (A2') may be a curved area that extends upward or downward from each of both left and right sides of the valve seat **300**.

[0077] As shown in FIG. **13**, a contact portion **330** that may contact the valve **20** may be formed in the valve seat **300**. The contact portion **330** may be a closed surface or a closed portion of the valve seat **300**. Here, the valve seat **300** may be formed to have a certain curvature to contact the outer peripheral surface of the valve **20**. Referring to FIGS. **13** and **15**, the valve seat **300** may be formed to be curved on both left and right sides with respect to a centerline (CL). Here, the curved curvature of the valve seat **300** and the curvature of the outer peripheral surface of the valve **20** may be formed to be identical or similar to each other.

[0078] A location indicator **302** may be formed on the top surface of the valve seat **300**. There is no restriction on the shape of the location indicator **302**. The valve seat **300** may be assembled to the valve housing **100** in a state in which the location indicator **302** is present on the upper side. That is, the location indicator **302** may prevent the valve seat **300** from being reverse-assembled to the valve housing **100**.

[0079] Referring to FIG. **14**, the valve seat **300** may include a first wall portion **360** and a second wall portion **370**. The first wall portion **360** and the second wall portion **370** may be formed to extend toward the rear of the valve seat **300**. The first wall portion **360** may be formed on the outer end of the valve seat **300**, and the second wall portion **370** may be formed inside the valve seat **300** at a certain distance from the first wall portion **360**. The first wall portion **360** may form the outer surface of the valve seat **300**, and the second wall portion **360** may be formed to surround the center hole **320**.

[0080] An insertion groove **380** may be formed between the first wall portion **360** and the second wall portion **370**. The insertion groove **380** may have a ring shape along the first wall portion **360** and the second wall portion **370**. The upper side of the insertion groove **380** may be open and the opposite side of its bottom may be the contact portion **330**.

[0081] A first groove **305** may be formed on the upper side of the valve seat **300**, and second grooves **307** and **308** may be formed on the lower side of the valve seat **300**. The first groove **305** and the second grooves **307** and **308** may be formed to extend along the front from the rear of the first wall portion **360** of the valve seat **300**. The number of first grooves **305** and the number of second grooves **307** and **308** may differ from each other. In another example embodiment, a single

first groove **305** may be formed and the second grooves **307** and **308** may be formed in two spaced apart from each other by a certain distance. However, there is no restriction on the number of first grooves **305** and the number of second grooves **307** and **308**.

[0082] A second fastening portion **350** may be formed on each of both left and right sides of the valve seat **300**. The second fastening portion **350** may be formed to extend along the vertical direction. The second fastening portion **350** may be formed in the first area (A1') of the valve seat **300**.

[0083] The second fastening portion **350** may include a second insertion shaped portion **350-1** and a second stopper shaped portion **350-2**. Description related to the shape of the second insertion shaped portion **350-1** and the second stopper shaped portion **350-2** may refer to contents described above in relation to the second insertion shaped portion **35-11** and the second stopper shaped portion **35-21** according to an example embodiment.

[0084] The second insertion shaped portion **350-1** and the second stopper shaped portion **350-2** may have the upper end surface and the lower end surface. Unlike an example embodiment, the lower end of the second insertion shaped portion **350-1** and the second stopper shaped portion **350-2** according to another example embodiment does not extend to the second area (A2').

[0085] Referring to FIG. **13**, the second stopper shaped portion **350-2** may have an inclined shape such that its one surface approaches the rear with getting closer from the upper side to the lower side. Viewing the side of the valve seat **300**, a distance (d1) from the upper side of the second stopper shaped portion **350-2** to the front may be less than a distance from the lower side of the second stopper shaped portion **350-2** to the front.

[0086] The second insertion shaped portion **350-1** and the second stopper shaped portion **350-2** may correspond to the aforementioned first insertion shaped portion **150-1** and first stopper shaped portion **150-2** of the first fastening portion **150**, respectively. Relationship between the second insertion shaped portion **350-1** and the second stopper shaped portion **350-2** and the first insertion shaped portion **150-1** and the first stopper shaped portion **150-2** of the first fastening portion **150** may refer to contents described in an example embodiment.

[0087] Meanwhile, the valve seat **300** according to another example embodiment may be manufactured using an injection method. Here, a pushing pin portion **310** through which an ejector pin operates may be formed in the valve seat **300**. The pushing pin portion **310** may have a shape that protrudes outward from the first wall portion **360**. A contact surface **311** through which the ejector pin may be contacted may be formed on the pushing pin portion **310**. The contact surface **311** may be formed on the rear surface of the valve seat **300**. The pushing pin portion **310** may be formed in the second area (A2') of the valve seat **300**.

[0088] Referring to FIG. **15**, a chamfer portion **365** may be formed in a portion in which an outer portion **360a** of the first wall portion **360** and an outer portion **330a** of the contact portion **330** are connected to each other, based on the horizontal cross-section of the valve seat **300**. The chamfer portion **365** refers to a portion in which an edge where two lines or surfaces meet each other at a certain angle is obliquely cut in a straight line. A first round portion **366** with a first curvature may be formed on the rear side of the chamfer portion **365**. The first round portion **366** may have the first curvature. The outer portion **360a** of the first wall portion **360** may be obliquely formed in the inner direction of the first wall portion **360** from the rear end of the first round portion **366** to one end of the second stopper shaped portion **350-2**.

[0089] A second round portion **367** may be formed in a portion in which an outer portion **370a** of the second wall portion **370** and the outer portion **330a** of the contact portion **330** are connected to each other, based on the horizontal cross-section of the valve seat **300**. The second round portion **367** may be formed with a second curvature.

[0090] The chamfer portion **365** and the first round portion **366** may be formed along the first wall portion **360**, and the second round portion **367** may be formed along the second wall portion **370**. Meanwhile, an area in which the chamfer portion **365** is formed may be larger than an area in

which the second round portion **367** is formed.

[0091] FIG. **16** illustrates a state in which a sealing portion is separated from a valve seat according to another example embodiment.

[0092] Referring to FIG. **16**, in another example embodiment, a sealing portion **400** may insert into the insertion groove **380** of the valve seat **300**. Description related to the sealing portion **400** may refer to the description made above in relation to the sealing portion **40** according to an example embodiment.

[0093] A first protrusion **405** may be formed on the upper side of the sealing portion **400**, and second protrusions **407** and **408** may be formed on the lower side thereof. The first protrusion **405** and the second protrusions **407** and **408** may be formed to protrude outward from the sealing portion **400**. The first protrusion **405** may have a shape that may insert into the first groove **305** of the valve seat **300**, and the second protrusions **407** and **408** may have a shape that may insert into the second grooves **307** and **308** of the valve seat **300**. The first protrusion **405** may be formed at a location corresponding to the first groove **305**, and the second protrusions **407** and **408** may be formed at locations corresponding to the second grooves **307** and **308**, respectively. The number of first protrusions **405** may be equal to the number of first grooves **305**, and the number of second protrusions **407** and **408** may be equal to the number of second grooves **307** and **308**. FIG. **17** is a view cut along line B-B' of FIG. **10**.

[0094] The principle that the second fastening portion **350** of the valve seat **300** fastens to the first fastening portion **150** of the valve housing **100** may refer to the contents described above in an example embodiment. As shown in FIG. **17**, referring to a state in which the valve seat **300** is fastened to the valve housing **100**, the valve seat **300** may be provided such that one side of the chamfer portion **365** of the valve seat **300** is almost adjacent to the first insertion shaped portion **150-1** of the first fastening portion **150**. The rear end of the chamfer portion **365** may be provided to face the curved shape of the first insertion shaped portion **150-1**.

[0095] Although the present invention is described with reference to specific example embodiments, it is provided as an example only and it will be apparent to one of ordinary skill in the art that various changes and modifications in forms and details may be made in these example embodiments without departing from the technical spirit of the various example embodiments. Therefore, other implementations, other example embodiments, and equivalents of the claims are to be construed as being included in the claims.

Claims

1. A valve seat assembly structure comprising: a valve housing including an inner wall and a single pair of first fastening portions having a shape that protrudes from the inner wall and provided to face each other; and a valve seat provided between the single pair of first fastening portions and including a second fastening portion formed on the side to couple with the first fastening portion, wherein each of the first fastening portion and the second fastening portion includes an insertion shaped portion having a curved shape and a stopper shaped portion having a planar shape.
2. The valve seat assembly structure of claim 1, wherein the first fastening portion includes: a (1-1)-th insertion shaped portion that vertically extends in the vertical direction of the valve housing; and a (1-1)-th stopper shaped portion that is formed in the inner direction of the first fastening portion at the end of the largest horizontal cross-section of the (1-1)-th insertion shaped portion.
3. The valve seat assembly structure of claim 2, wherein the first fastening portion includes: a (1-2)-th insertion shaped portion that extends in a curved shape from the lower side of the (1-1)-th insertion shaped portion; and a (1-2)-th stopper shaped portion that extends in the inner direction of the first fastening portion at the end of the largest horizontal cross-section of the (1-2)-th insertion shaped portion.
4. The valve seat assembly structure of claim 1, wherein the second fastening portion includes: a

(2-1)-th insertion shaped portion that is formed in a straight area on the side of the valve seat, and vertically extends in the vertical direction of the valve seat; and a (2-1)-th stopper shaped portion that is formed in the inner direction of the second fastening portion at the end of the largest horizontal cross-section of the (2-1)-th insertion shaped portion.

5. The valve seat assembly structure of claim 4, wherein the second fastening portion includes: a (2-2)-th insertion shaped portion that is formed in a curved area on the side of the valve seat and extends in a curved shape from the lower side of the (2-1)-th insertion shaped portion; and a (2-2)-th stopper shaped portion that extends in the inner direction of the second fastening portion at the end of the largest horizontal cross-section of the (2-1)-th insertion shaped portion.

6. The valve seat assembly structure of claim 1, wherein the valve seat is curved on both left and right sides to have a constant curvature based on a vertical centerline.

7. The valve seat assembly structure of claim 1, wherein the valve seat includes: a first wall portion formed on its outside; and a second wall portion formed to surround a center hole of the valve seat, and an insertion groove is formed between the first wall portion and the second wall portion.

8. The valve seat assembly structure of claim 7, wherein a sealing portion inserts into the insertion groove.

9. The valve seat assembly structure of claim 8, wherein the sealing portion is an X-ring or an O-ring.

10. A valve seat assembly structure comprising: a valve housing including an inner wall and a single pair of first fastening portions having a shape that protrudes from the inner wall and provided to face each other; a valve seat formed with an insertion groove, provided between the single pair of first fastening portions, and having a second fastening portion to couple with the first fastening portion; and a sealing portion configured to insert into the insertion groove, wherein the valve seat includes a closed portion, a first wall portion formed on the outside based on the insertion groove, and a second wall portion formed on the inner side than the first wall portion, a chamfer portion obliquely provided to face the first fastening portion is formed in a portion in which an outer portion of the closed portion and an outer portion of the first wall portion are connected to each other, and a round portion having a first curvature is formed in a portion in which the outer portion of the closed portion and an outer portion of the second wall portion are connected to each other.
